

SIYUAN LIU

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EDUCATION

BSc in Computer Science, The Hong Kong Polytechnic University Sep. 2023 – Jun. 2027 (Expected)
Minor in Applied Mathematics • Dean's Honours List, 2024/25

Summer School Program in Computer Science, Stanford University, Palo Alto, US Jun. – Aug. 2025

Exchange Program in Math, University of Waterloo, Waterloo, Canada Sep. – Dec. 2025

RESEARCH

Hierarchical TTT: Efficient Test-Time Learning through Trainable Guidance Jun. 2026 – Present
MIT Media Lab — Multisensory Intelligence Group Supervisor: Prof. Paul Pu Liang

- Built a **Hierarchical Test-Time Training (TTT)** framework using **slime** for efficient test-time adaptation of language models on frontier-level CS tasks.
- Designed a reinforcement learning-based search mechanism with **PUCT sampling** and top- k node attachment strategies to optimize guidance generation and improve iterative reasoning performance.
- Developed prompt designs for the trainable **guidance model**, enabling high-level strategic planning while keeping the execution model frozen; achieved **state-of-the-art performance on TriMul** task.

A*-Thought-V2: Efficient Chain-of-Thought Compression via LLM Dynamics Jan. 2026 – May. 2026
Tsinghua NLP Lab Supervisor: Prof. Shuo Wang

- Proposed a **PCA-based reasoning trajectory compression method** that projects LLM hidden states into a low-dimensional space to identify redundant reasoning steps, enabling efficient Chain-of-Thought compression while improving reasoning performance.
- Established experimental baselines for efficient reasoning approaches in LLMs and conducted comprehensive evaluations to analyze the trade-off between reasoning accuracy and inference efficiency.

ReaLLM-Med-Eval — A Robust Evaluation Framework for Medical LLMs Jan. 2025 – Jun. 2025
ReaLLM-Labs, Department of Computing, PolyU Supervisor: Prof. Hongxia Yang

- Leveraged **VLMEvalKit** to conduct systematic evaluations of multimodal medical LLMs on MLLM benchmarks, analyzing model capabilities across diverse medical reasoning tasks.
- Engineered high-quality medical instruction datasets to support SFT and reproducible evaluation, contributing to [arXiv:2505.23867](#) through experiments, analysis, and visualization.

INTERNSHIP

Research Intern, ModelBest Inc. Dec. 2025 – Apr. 2026

- Conducted systematic evaluations of 8B–14B LLMs by extending OpenCompass and leveraging Evalchemy for validation, analyzing model capabilities and limitations across extended-context scenarios.
- Designed an SFT data generation pipeline to improve long-context capabilities by extracting entity anchors from classic literature and systematically constructing targeted questions around them.

PUBLICATIONS

MiniCPM-SALA: Hybridizing Sparse and Linear Attention for Efficient Long-Context Modeling Feb. 2026
arXiv preprint ([arXiv:2602.11761](#))

MiniCPM Team, Wenhao An, Yingfa Chen, Yingfa Chen, Yewei Fang, ..., **Siyuan Liu**, ..., Xu Han, Zhiyuan Liu, Maosong Sun

Infi-Med: Low-Resource Medical MLLMs with Robust Reasoning Evaluation

May. 2025

arXiv preprint ([arXiv:2505.23867](#))

Zeyu Liu, Zhitian Hou, Yining Di, Kejing Yang, Zhijie Sang, Congkai Xie, Jingwen Yang, **Siyuan Liu**, Jialu Wang, Chunming Li, Ming Li, Hongxia Yang

Enhancing Suicide Risk Detection with MDF-FOF

Jun. 2024

IEEE International Conference on Big Data (IEEE BigData 2024)

Zheng Shouwen, Zhou Taiqi, Tao Yingzhi, Chen Junru, Wang Ruofei, **Siyuan Liu**

SKILLS

Tools

Codex (/plan, /goal), Claude Science(research workflow automation)

LLM Training

Large-scale model training, RL (verl, slime), SFT (LLaMA-Factory)

AWARDS

- **IEEE BigData 2024 Cup** — Team ranked **1st out of 47 teams** with cash prize

2024
- **Talent Development Scholarship**, HKSAR Government Scholarship Fund (HK\$10,000)

2025, 2026
- **Most Outstanding Academic Achievement Award** — Recognised for top academic performance

2024, 2025
- **Dean’s Honours List** — Recognised for outstanding academic performance

2024/25

EXTRA-CURRICULAR ACTIVITIES

- **Chapter President**, Enactus PolyU, Hong Kong Polytechnic University

Aug. 2024 – Aug. 2025
- **Champion**, USFHK Women Soccer

2024, 2025